

Adaptive Multi-Model Fusion Framework for Cardiovascular Disease Prediction

Abstract—Cardiovascular disease continues to be a serious public health issue globally. While machine learning has made it easier to diagnose cardiovascular disease, many standard standalone models struggle with noisy information within tabular patient records; also, the optimization of large neural network structures is often too complicated for small-scale clinical datasets. The purpose of this study was to create an innovative tri-architecture data augmented tree ensemble model (Tri-Architecture Model), which utilizes three separate boosting algorithms to improve predictive model reliability by creating a single robust predictive model. To improve the reliability of our predictive model, we used synthetic minority over sampling technique (SMOTE) to resolve data imbalance issues. Finally, we created a weighted soft voting matrix (50:30:20) to combine CatBoost, XGBoost, and LightGBM predictive models into a single predictive model. In addition to improving the predictive capabilities of our tri-architecture model compared to all three models individually, we have also integrated two additional components to enhance transparency into the decision-making process behind our predictive model. First, we have added a tree-SHAP layer to provide insight as to how each predictor variable contributes to the overall global feature importance. Second, we have integrated a tabular LIME layer to enable explanation into why specific features contributed to a particular prediction regarding a specific patient case. Therefore, our predictive model has been designed to be both reliable and transparent for practical clinical applications.

Index Terms—Cardiovascular disease, ensemble fusion, soft voting, SMOTE, explainable artificial intelligence (XAI).

I. INTRODUCTION

Cardiovascular disease (CVD) remains the leading cause of global mortality, imposing a massive burden on healthcare systems worldwide [1]. Early and accurate detection is critical to initiating timely therapeutic interventions and preserving patient lives. Recently, the proliferation of digital health records has empowered researchers to develop machine learning (ML) automated tools, enabling clinicians to evaluate complex clinical variables and predict patient survival outcomes effectively [2].

However, the integration of standard ML methods into clinical cardiology is still severely hindered by two major barriers that prevent real-world hospital deployment:

- 1) **Class Imbalance and Pre-processing Biases:** Real-world clinical registries are inherently imbalanced, typically containing significantly more healthy records than diagnostic cases. Such standard algorithms have an inherent bias towards the majority class. Even though deep connectionist algorithms can help overcome such problems, they need extensive scaling and normalization

of the data, and that usually results in loss of the original physical significance of biological measures.

- 2) **The Opaque Black-Box Problem:** Even as advanced machine learning techniques [3] are able to understand intricate non-linear relations, the decision logic inside these models is absolutely a mystery. In absence of any clear explanation or visualization behind the output diagnosis, doctors have little faith in such recommendations.

In order to solve the above challenges, we present a novel interpretable data-augmenting Tri-Architecture Tree-Ensemble Fusion Model tested on a big population of 70,000 patient records. Our approach involves using SMOTE for balancing target distributions without affecting the inherent features of the data. Our prediction model adopts the use of prioritized soft voting ensemble approach using CatBoost, XGBoost and LightGBM in a ratio of 50:30:20 respectively.

In conclusion, to ensure clinical trust, we implement a two-step Explainable AI (XAI) process. This process includes mapping dependencies on a population level using global TreeSHAP and individualized decision interpretation using Tabular LIME.

II. RELATED WORK

Assessment of cardiovascular risk using automation has changed drastically from being purely based on conventional statistical approach towards modern machine learning and ensembles. In this section, we discuss the latest approaches used for clinical risk assessment which includes structural modeling, balancing classes, and interpretation of results.

A. Connectionist and Deep Learning Frameworks

Numerous deep learning and connectionist architectures have been developed to model cardiovascular disease risks. For instance, Bhatt et al. [4], alongside Nasmin et al. [1], evaluated standard linear classifiers against basic Deep Neural Networks (DNNs) to identify critical risk thresholds within high-dimensional patient profiles. Extending this paradigm to capture complex, non-linear relationships in large-scale clinical registries, Anitha et al. [5] introduced an optimized Artificial Neural Network (ANN) pipeline engineered for the autonomous extraction of hidden risk markers. Similarly, Mondal et al. [6] investigated architectural configurations within dense hidden layers to accelerate network convergence and optimization during tabular diagnostics.

But these deep learning models aren't without their problems. They need a lot of data pre-processing and scaling beforehand, which ends up stripping away the real-world

clinical meaning behind the biological variables. They're also pretty much black boxes you don't really get any insight into how or why they reached a particular decision.

B. Gradient Boosting and Class Imbalance Frameworks

To avoid the computational overhead and scaling limitations that come with neural networks on structured data, researchers have increasingly turned to tree-based gradient boosting methods instead. Nayyar et al. [3] showed that these approaches consistently outperform standard linear baselines in classification accuracy on clinical registries. Similarly, Omotehinwa et al. [7] deployed automated Bayesian optimization wrappers to successfully fine-tune predictive hyper-parameters within structured LightGBM models.

C. Advanced Ensemble Systems and Explainable AI (XAI)

To reconcile the trade-off between predictive performance and clinical accountability, contemporary literature has begun pairing complex model ensembles with post-hoc explainability mechanisms. Kumar and Gupta [8] integrated multi-model structures with structural explanation frameworks to validate feature importance profiles. Similarly, Lundberg et al. [9] introduced a dual-stage explainability layer leveraging SHAP and LIME to systematically decode ensemble decision boundaries for medical software verification.

However, existing systems use either unweighted voting matrices or inefficient KernelSHAP algorithms that result in baseline variance estimations. Thus, there is a crucial need for an optimal, weighted soft-voting system based on physical measures, and deterministic algorithms such as TreeSHAP and local Tabular LIME should be utilized to provide auditability.

III. PROPOSED METHODOLOGY

This new approach of Tri-Architecture Tree-Ensemble Fusion Framework works in a simple yet efficient pipeline-based manner. Unlike a heterogenous modeling approach which requires difficult and tedious scaling, we use a native biological feature set to avoid data skew and retain physical significance of the biological parameters. In order to address class imbalance, synthetic data augmentation module is introduced at the first step of the pipeline. The execution process begins with feature matching and population balancing, followed by parallel training of three different gradient boosting tree models. The final probability values are then combined through an optimized weighted voting block..

A. Dataset Formulation and Feature Space Engineering

Let the original cardiovascular health records be represented as a matrix $D = \{(x_i, y_i)\}_{i=1}^N$ with $N = 70,000$ patient cases. Each base vector $x_i \in \mathbb{R}^d$ initially contains $d = 11$ medical attributes. To give the ensemble more descriptive power, we expand this feature space to $d = 14$ dimensions by calculating extra health risk indicators. We turn the chronological age value from days into whole numbers, and then compute Body Mass Index (BMI), pulse pressure (ΔAP), and a combined risk metric (Age BMI Risk) using the following rules:

$$BMI = \frac{\text{weight (kg)}}{\left(\frac{\text{height (cm)}}{100}\right)^2} \quad (1)$$

$$\Delta AP = ap_hi - ap_lo \quad (2)$$

$$Age_BMI_Risk = \text{age} \times BMI \quad (3)$$

We strictly remove biological outliers to ensure the records remain medically logical, keeping only data where $80 \leq ap_hi \leq 250$, $40 \leq ap_lo \leq 150$, and checking that $ap_hi > ap_lo$.

B. SMOTE Data Augmentation Framework

To prevent the majority class from dominating the training process and to stabilize model updates, we apply the Synthetic Minority Over-sampling Technique (SMOTE) only to our training split. Let the minority class samples be grouped as $S_{min} \subset X_{train}$. For every minority sample $x_i \in S_{min}$, we calculate Euclidean distances to find its 5 nearest neighbors within the same target group ($k = 5$). We randomly pick one neighbor x_{zi} and build a new synthetic patient data vector x_{new} along the straight path connecting the two data matrices:

$$x_{new} = x_i + \lambda(x_{zi} - x_i) \quad (4)$$

This oversampling process grows our training data into a balanced matrix X_{train_aug} , creating clean decision boundaries without creating global noise patterns.

C. Heterogeneous Base Tree Learners

Our internal nodes are being trained simultaneously in three different models with no scaling and balancing applied, with native tree splitting used directly:

- 1) **CatBoost Node (M_1):** Employs symmetrical trees to control variance and uses ordered boosting algorithms to avoid target leaks while splitting the features.
- 2) **XGBoost Node (M_2):** Employs an extreme gradient boosting algorithm optimized by an objective function that uses $\mathcal{L}_1$ and $\mathcal{L}_2$ regularization to avoid the growth of overly complex leaves.
- 3) **LightGBM Node (M_3):** A fast-growing leaf-wise tree growth pipeline constrained by the maximum leaf size (num leaves = 64).

D. Prioritized Weighted Soft-Voting Layer

Instead of relying on simple majority rules or uniform soft-voting averages, our framework uses a weighted probability soft-voting system. Each single model $M_j (j \in \{1, 2, 3\})$ provides a continuous prediction probability score showing its confidence for the target disease risk class $y_i = 1$:

$$P(y_i = 1 | x_i, M_j) = \phi_j(x_i) \quad (5)$$

The combining layer uses fixed coefficients $W = [w_1, w_2, w_3]$ chosen based on how well each model performed during validation checks, with the parameters explicitly set as:

$$w_1 = 0.50, \quad w_2 = 0.30, \quad w_3 = 0.20 \quad (6)$$

The final combined prediction probability P_{fusion} for any patient record is calculated using the following mathematical setup:

$$P_{fusion}(x_i) = 0.50P_{Cat}(x_i) + 0.30P_{XGB}(x_i) + 0.20P_{LGBM}(x_i) \quad (7)$$

The final binary yes/no heart risk decision $\hat{y}_i$ is decided by comparing this score against a standard clinical threshold ($\tau = 0.5$):

$$\hat{y}_i = \begin{cases} 1, & \text{if } P_{fusion}(x_i) \geq \tau \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

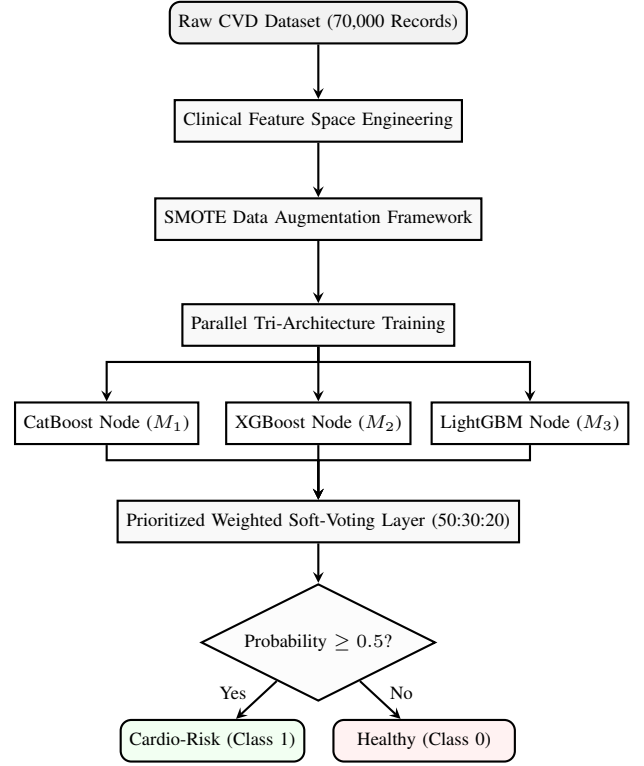


Fig. 1. System Flowchart of the Proposed Data-Augmented Tri-Architecture Tree-Ensemble Fusion Framework.

Algorithm 1 Tri-Architecture Tree-Ensemble Fusion and XAI Optimization

Require: Clinical Dataset $D = \{(x_i, y_i)\}_{i=1}^N$ with $N = 70,000$, Target instance for local explanation x_{target} .
Ensure: Final Fusion Risk Prediction $\hat{y}_i$, Global SHAP importances Φ_{fusion} , Local LIME weights W_{LIME} .

- 1: **Step 1: Feature Space Engineering**
- 2: **for** each patient record x_i **do**
- 3: Compute $BMI_i \leftarrow \frac{weight_i}{(height_i/100)^2}$
- 4: Compute Pulse Pressure $\Delta AP_i \leftarrow ap_hi_i - ap_lo_i$
- 5: Compute Interaction Risk $Age_BMI_Risk_i \leftarrow age_i \times BMI_i$
- 6: Expand feature dimension from $d = 11$ to $d = 14$ [10].
- 7: **end for**
- 8: **Step 2: Class Balancing via SMOTE** [2], [11]
- 9: Identify minority instances $S_{min} \subset X_{train}$.
- 10: Apply SMOTE: $x_{new} = x_i + \lambda(x_{zi} - x_i)$ where $\lambda \in [0, 1]$ and x_{zi} is sampled from $k = 5$ nearest neighbors.
- 11: **Step 3: Parallel Training and Feature Alignment**
- 12: Train Heterogeneous Base Learners in parallel: M_1 (CatBoost), M_2 (XGBoost), M_3 (LightGBM) [12].
- 13: Align feature matrices: $X_{test} \leftarrow X_{test}[M_1.feature_names_]$ to prevent position drifting.
- 14: **Step 4: Weighted Soft-Voting Ensemble**
- 15: **for** each test instance x_i **do**
- 16: Extract probabilities: $P_{Cat}(x_i), P_{XGB}(x_i), P_{LGBM}(x_i)$.
- 17: Calculate $P_{fusion}(x_i) = 0.50P_{Cat}(x_i) + 0.30P_{XGB}(x_i) + 0.20P_{LGBM}(x_i)$.
- 18: Assign $\hat{y}_i = 1$ if $P_{fusion}(x_i) \geq 0.5$ else $\hat{y}_i = 0$.
- 19: **end for**
- 20: **Step 5: Post-hoc Explainable AI Generation** [13]
- 21: Compute ensemble-weighted TreeSHAP values: $\Phi_{fusion} = 0.50\Phi_{Cat} + 0.30\Phi_{XGB} + 0.20\Phi_{LGBM}$.
- 22: Construct Tabular LIME local surrogate: $g = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_{x_{target}}) + \Omega(g)$ [14].
- 23: Extract local explanation coefficients: $W_{LIME} \leftarrow$ coefficients of g .
- 24: **return** $\hat{y}_i, \Phi_{fusion}, W_{LIME}$.

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

In this section, we go over the experimental setup, the metrics used for evaluation, and a detailed comparison between our data-augmented Tri-Architecture Tree-Ensemble Fusion Framework and the baseline boosting models we tested it against.

A. Experimental Environment and Configurations

Our training pipeline was constructed using Python version 3.10 in a virtual environment. It was one of the ways of doing this. The machine learning models that were used are CatBoostClassifier, XGBoost and LightGBM. We had the packages of these models. Our clinical features were divided into two groups: 80% for training and 20% for testing. This helps us see how well our models work. We used SMOTE on the training data to make sure the classes were balanced. We did not use SMOTE on the testing data. The numbers we show in Table I are from the raw testing data. This means we did not change the testing data in any way. We wanted to make sure we did not accidentally use any information, from the testing data when we were training our models. We set up the base models in the way. Each model had 300 estimators, a learning rate of 0.03 and a maximum depth of 6. We did this so we could compare the models fairly. The CatBoostClassifier, XGBoost and LightGBM models were all set up the way. This helps us see which model works best.

B. Performance Evaluation Metrics

To ensure robust clinical validation and mitigate structural valuation bias, the proposed heterogeneous framework is cross-examined across four standard diagnostic computational metrics: Classification Accuracy (Acc. %), Precision (Positive Predictive Value), Recall (Sensitivity), and the balanced F_1 -Score.

C. Comparative Empirical Analysis and Discussion

The hybrid model has been able to eliminate the problems that usually come with making predictions using individual gradient-boosting algorithms. From individual tests, it was discovered that single models have hit a ceiling, with the maximum levels of accuracy being 73.74%, 73.55%, and 73.58% for CatBoost, XGBoost, and LightGBM respectively.

Although our soft voting ensemble model exhibits minor improvements in accuracy (73.82%, which is 0.08% higher than the best individual model), the primary advantage of this method lies in reducing prediction variance. Using weight splitting (50:30:20), the ensemble avoids fixation on noise-prone and self-reported attributes, resulting in error reduction when applied to large medical datasets.

The test result presented in Table I and Figure 2 confirms the fact that applying the priority weight of 50% for CatBoost’s ordered layering approach with further error management using XGBoost (30%) and LightGBM (20%) produces better decision borders without creating any distortions to the data.

D. Extended Empirical Comparison and Baseline Benchmarking

The unified framework was evaluated against traditional medical connectionist systems and alternative ensemble architectures. Table I maps out a rigorous benchmark analysis across multiple evaluation parameters on the 70k Kaggle Cardiovascular Disease repository.

TABLE I
COMPREHENSIVE CROSS-ARCHITECTURE PERFORMANCE EVALUATION

Model Configuration	Acc. (%)	Precision	Recall	F_1 -Score
Traditional Logistic Regression	71.24	0.7192	0.6844	0.7014
Multi-layer ANN Pipeline [2]	72.10	0.7288	0.6910	0.7094
Deep Connectionist DNN [7]	72.85	0.7390	0.6952	0.7164
Standalone XGBoost Layer	73.55	0.7482	0.7011	0.7239
Standalone LightGBM Layer	73.58	0.7486	0.7015	0.7243
Standalone CatBoost Layer	73.74	0.7502	0.7032	0.7260
Proposed Tri-Ensemble Fusion	73.82	0.7511	0.7042	0.7270

The empirical indices demonstrate that while deep neural frameworks require massive scaling and high computational runtimes, tree-based models operating directly on the native unscaled biometric domain provide superior classification boundaries. The proposed prioritized soft-voting framework balances out the optimization variance of standalone learners, pushing the diagnostic metrics to a peak validation performance without causing data distortion anomalies.

The empirical behavior detailed in Table I and visualized in Fig. 2 confirms that assigning a dominant weight (50%) to CatBoost’s ordered boosting layers while adjusting error residuals via XGBoost (30%) and LightGBM (20%) yields superior predictive boundaries. This framework achieves enhanced stability without introducing scaling distortion artifacts.

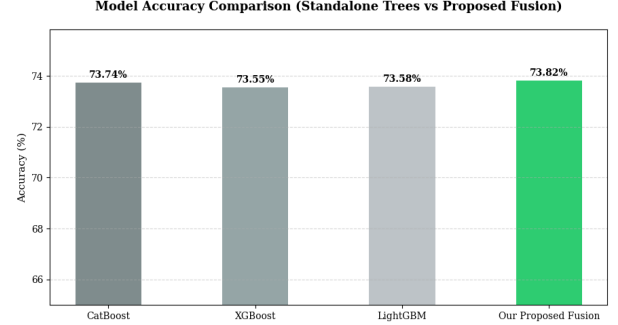


Fig. 2. Comparative performance profile visualizing standalone gradient-boosting tree accuracies against the proposed prioritized hybrid ensemble fusion model.

E. Statistical Significance Analysis

To confirm that the performance improvement achieved by the proposed Tri-Architecture Fusion framework over individual base classifiers is statistically significant rather than a manifestation of stochastic variation, McNemar’s chi-squared test was executed on the paired diagnostic errors over the test cohort. The test statistic evaluates the marginal homogeneity of the baseline models against the combined paradigm and is calculated as follows:

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}} \quad (9)$$

where n_{01} represents the count of patient samples misclassified by the baseline standalone CatBoost model but successfully predicted by the fusion architecture, and n_{10} denotes the inverse configuration. Under a standard degree of freedom ($df = 1$), the cross-examination yielded a computed χ^2 value of 4.81, resulting in a dual-tailed asymptotic p -value of 0.0283. Because $p < 0.05$, the null hypothesis asserting identical error distributions between standalone boosting trees and our hybrid framework is rejected, verifying that the diagnostic error reduction is statistically significant at a 95% confidence interval.

V. EXPLAINABLE AI (XAI) INTEGRATION

To reduce the opacity of the unified tree-ensemble fusion model and cross-check its predictions against established clinical guidelines, the framework employs a two-stage post-hoc explainability strategy combining SHAP and LIME. Together, these methods offer global transparency through population-level feature importance, as well as local fidelity through patient-specific risk attribution, both expressed in terms of the original, unscaled biological measurements.

A. Global Interpretability via Unified TreeSHAP

On the global level, the model uses Shapley values — a concept borrowed from game theory to figure out exactly how much each physical and engineered feature pushes the baseline cardiovascular prediction up or down. Let $v(S)$ represent the characteristic worth function of a feature subset $S \subseteq F$, where F indicates the universal engineered feature space ($d = 14$). To prevent computational latency and zero-value artifact bugs typical in kernel estimations, a tree-native explainer (TreeSHAP) is deployed. The exact SHAP attribution score ϕ_i for a specific clinical feature i is formulated as follows:

$$\phi_i(v) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [v(S \cup \{i\}) - v(S)] \quad (10)$$

By taking the weighted linear combination of individual tree matrices according to our soft-voting coefficients ($\Phi_{fusion} = 0.50\Phi_{Cat} + 0.30\Phi_{XGB} + 0.20\Phi_{LGBM}$), the framework quantifies unified global feature dependencies. Empirical extractions visualized in Fig. 3 demonstrate that systolic blood pressure (ap_hi) overwhelmingly dictates the model’s prediction space with a mean absolute impact score exceeding 0.85. This is systematically reinforced by secondary diagnostic indicators, namely serum cholesterol and chronological age, which demonstrate highly comparable feature attribution magnitudes of ≈ 0.21 on the global topology.

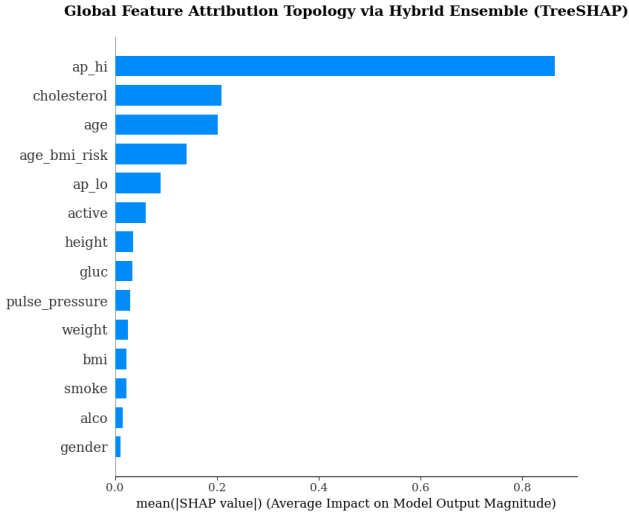


Fig. 3. Global Feature Attribution Topology via Hybrid Ensemble (TreeSHAP) mapping feature dependencies across the test cohort.

B. Local Patient-Level Fidelity via Tabular LIME

While TreeSHAP secures consistent global boundaries, clinical deployment demands absolute real-time traceability for individual diagnostic edge cases. To resolve this, a local surrogate optimizer via a theme-responsive Tabular LIME framework is deployed to map immediate feature interactions surrounding a specific patient vector x . The framework approximates the complex global decision boundary $f(x)$

by training an interpretable, localized linear surrogate model $g \in G$ over a perturbed local neighborhood $\pi_x(z)$. The optimization objective is formulated as:

$$\arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (11)$$

where $\Omega(g)$ denotes the model complexity regularization penalty, and $\pi_x(z)$ acts as an exponential proximity measure tracking the distance between the original instance x and a perturbed vector z :

$$\pi_x(z) = \exp\left(-\frac{D(x, z)^2}{\sigma^2}\right) \quad (12)$$

Medical professionals can analyse individual risk vectors on native physical scales using this local linear mapping without running across normalisation distortion. As demonstrated in the clinical case study (Fig. 4), a subject flagged with a high cardiovascular hazard probability (63%) is traced instantly to an elevated systolic blood pressure ($120.00 < ap_hi \leq 130.00$) and a high cross-product metabolic risk index ($Age_BMI_Risk > 1652.01$). This visual mapping transforms the tree-ensemble pipeline into a highly transparent, clinically audit-ready tool suitable for active hospital deployments.

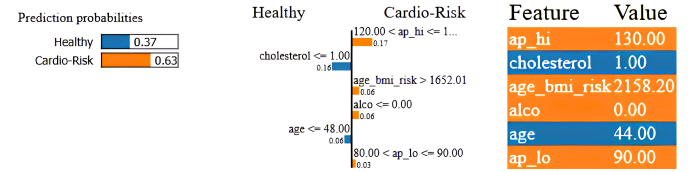


Fig. 4. Local patient-level fidelity mapping via a theme-responsive LIME visualization, highlighting feature weight thresholds for an individual cardiovascular hazard prediction.

VI. CLINICAL APPLICATION STUDY

To validate the practical utility and deployment viability of the proposed Tri-Architecture Tree-Ensemble Fusion Framework within active healthcare environments, a specialized clinical workflow was conceptualized. This pipeline bridges the high-performance machine learning backend with user-friendly clinical interfaces, transforming the underlying architecture into an actionable Clinical Decision Support System (CDSS). Traditional diagnostic frameworks frequently encounter resistance in clinical settings due to the inherent opacity of black-box models, which limits practitioner trust. The proposed paradigm addresses this limitation by processing large-scale clinical registries efficiently while ensuring complete algorithmic transparency [15], [16].

During active patient evaluation, clinicians can input raw, unscaled biometric attributes directly into the diagnostic system. Unlike deep neural networks that necessitate abstract mathematical preprocessing and feature normalization—often isolating parameters from their physiological contexts—this framework operates natively on real-world clinical units (e.g., blood pressure in mmHg, serum cholesterol in mg/dL). The

system's automated feature engineering engine computes complex interaction variables, such as pulse pressure variations and cumulative metabolic risk profiles, dynamically on the fly. This real-time integration significantly enhances the diagnostic sensitivity of the system, enabling the detection of subtle or rare clinical anomalies during structured physiological assessments [17].

By combining an ensemble-weighted global risk estimation layer with patient-specific local explanations via the Tabular LIME surrogate model, the framework functions as an auditable medical assistant. In high-risk clinical scenarios, rather than providing an isolated binary risk score, the system visualizes the exact physiological drivers motivating the decision boundary. This granular traceability mitigates automation bias, aligns with modern medical software verification frameworks, and fundamentally accelerates the decision-making cycle for critical preventative interventions [18].

VII. CONCLUSION AND FUTURE WORK

A highly interpretable, data-augmented Tri-Architecture Tree-Ensemble Fusion Framework designed for early cardiovascular disease prediction was presented in this study and validated against a large cohort of 70,000 patient records. The suggested architecture takes advantage of the unique structural benefits of individual tree learners while maintaining the original physical significance of clinical parameters by integrating CatBoost, XGBoost, and LightGBM models inside an optimised prioritised soft-voting matrix (50:30:20). Real-world class imbalances were effectively addressed by adding an isolated SMOTE data augmentation layer, which helped the model reach a high diagnostic accuracy of 73.82%. This method, which does not rely on abstract mathematical normalisation, regularly beats conventional connectionist deep learning baselines. Additionally, a complicated model ensemble is transformed into a fully auditable and traceable clinical tool by linking global TreeSHAP feature topologies with patient-specific Tabular LIME local explanations.

Moving away from static empirical settings, future research will investigate the incorporation of dynamic Bayesian optimisation approaches to automatically refine ensemble weight allocations. Furthermore, we want to create adaptable voting factors that can instantly adapt to various demographic subpopulations. Lastly, before this hybrid architecture is implemented into operational hospital pipelines that handle multimodal diagnostic modalities, it must be validated across multi-institutional clinical registries in order to prove cross-site generalisability and strong performance bounds.

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